MILITARY SPECIFICATION

CIRCLE, AZIMUTH (MARK 4 MOD 1) AND CIRCLE, BEARING (MARK 2 MOD 1)

FOR 6-INCH INDICATORS

1. SCOPE

1.1 This specification covers azimuth and bearing circles, used with Mark 3 (6-inch card) transmitting magnetic compass (MIL-C-23472) and type "B" (6-inch card) ship's course indicators (MIL-I-15676), for accurate determination of the azimuths of celestial bodies and bearings of terrestrial objects.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/8 - Aluminum Alloy Bar, Rod, Shapes, and Tube, Extruded, 6061 and 6062.

MILITARY

MIL-G-174 - Glass, Optical.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements For.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-O-16898 - Optical Elements; Packaging of.

STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

MILITARY

S2407-533624 - Six Inch Compass - Top Bezel Details.
400-533991 - Circle, Azimuth, MK 4 MOD 1 Assembly.
400-533992 - Circle, Azimuth, MK 4 MOD 1 Details.
400-533993 - Circle, Azimuth, MK 4 MOD 1 Details.
400-533994 - Circle, Azimuth, MK 4 MOD 1 Details.
400-533995 - Circle, Bearing, MK 2 MOD 1 Assembly.
400-533996 - Circle, Bearing, MK 2 MOD 1 Details.
400-533997 - Circle, Bearing, MK 2 MOD 1 Details.
400-533998 - Case, Carrying for MK 4 MOD 1 Azimuth Circle and MK 2 MOD 1 Bearing Circle.

PUBLICATIONS

MILITARY

NAVSHIPS 0924-004-7000 - Manual for Maintenance and Repair of 6 Inch Azimuth Circle, MK 4 MOD 1; 6 Inch Bearing Circle, MK 2 MOD 1.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6605

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction model.- Unless otherwise specified (see 6.2), and as soon as practicable after award of the contract or order and prior to submission of any complete units for final acceptance, the contractor shall furnish a preproduction model for inspection (see 4.2) to determine conformance with this specification (see 6.3). Approval of the preproduction model by the command or agency concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any deviations in production units from the approved preproduction model shall be subject to the written approval of the command or agency concerned.

3.2 Description.-

3.2.1 Azimuth circle.- The azimuth circle shall consist of a nonmagnetic, metal ring, equipped with sights, which can be mounted on the bezel of a 6-inch ship's course indicator for visual determination of bearing or azimuth. The circles are equipped with two pairs of sight assemblies located perpendicularly to each other. The front sight, consisting of a black mirror, open sight vane and pentaprism housing is lined up with the rear vane for the sighting of either celestial or terrestrial objects. The second pair of sights consists of a cylindrical mirror and a right angle prism assembly. The cylindrical mirror receives the sun's image and projects it into the slit of the right angle prism; upon passing through the assembly, the image is projected onto the card of the indicator. For proper positioning each pair of sights is equipped with a level. Two locking knobs, mounted opposite each other on the ring surface, permit adjustment to the desired angle and subsequent locking at the final desired angle.

3.2.2 Bearing circle.- The bearing circle differs only in minor details from the azimuth circle. Its construction, based on the same design, employs the identical components, except for the black mirror, the concave mirror and right angle prism assembly, (see figures 1 and 2).

3.3 Construction.- The azimuth and bearing circles shall be so constructed that vibration, shock and extremes of temperature, which may be encountered in service aboard Naval ships, will not adversely affect their performance.

3.3.1 The azimuth circle shall be constructed in accordance with Drawings 400-533991 through 400-533994. The bearing circle shall be constructed in accordance with Drawings 400-533995 through 400-533997. The carrying case for either the azimuth or bearing circle shall be constructed in accordance with Drawing 400-533998. Figures 1 and 2 show the general arrangement of the azimuth circle and bearing circle.

3.3.2 Azimuth and bearing circle rings.- The azimuth and bearing circle rings shall be made of aluminum conforming to QQ-A-200/8, temper T6. The rings shall be turned true to fit over the bezel of a Mark 3 magnetic compass or a type "B" ship's course indicator shown on Drawing S2407-533624.

3.4 Materials.-

3.4.1 Nonmagnetic.- Since the azimuth and bearing circles are to be used on the bezel of a magnetic compass, all materials used in the construction of the azimuth and bearing circles shall be nonmagnetic (having a magnetic permeability of 1.02 or less after fabrication).

3.4.2 Glass.- Optical elements shall conform to MIL-G-174. Elements shall be free from strain, seeds, stones, striae and bubbles. Polished surfaces shall be free from objectionable scratches, pits, stains and other defects visible in the field of view or which would impair performance.

3.4.3 Lubricants and sealing materials.- All lubricants and sealing materials shall be suitable for their intended purpose throughout the temperature range specified in 3.5, and shall not cause or contribute to filming of the optical surfaces, or support the growth of fungi.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

3.5 Temperature and humidity.- The assembled azimuth or bearing circle shall meet the requirements of extreme temperature for exposed locations as specified in MIL-I-983.

3.6 Salt spray.- The assembled azimuth and bearing circle shall show no appreciable corrosion or permanent malfunctioning of rotatable, movable or removable parts when subjected to the salt spray test specified in MIL-I-983.

3.7 Shock and vibration.- The assembled azimuth or bearing circle, when mounted on the bezel (or facsimile) of a Mark 3 magnetic compass or the type "B" ship's course indicator, shall satisfactorily withstand, without damage which would cause malfunctioning or inoperation, type I vibration tests in accordance with MIL-STD-167 and type A HI shock in accordance with MIL-S-901.

3.8 Alignment.- All parts shall be assembled so as to remain in alignment when tested as specified in 4.4.4. The assembled parts shall be in a plane perpendicular to the plane of the bearing ring when the instrument is level. The image of the sun reflected from the cylindrical mirror through the right angle prism shall cast a well defined pencil of light (0.050 inch maximum width between 0 and 2-3/8 inch below mounting surface of circle) on the compass card when the azimuth circle is level. Spirit level assemblies shall be adjusted to indicate the instrument is level to within plus or minus 10 minutes.

3.9 Carrying case.- Each azimuth or bearing circle shall be furnished in a stowage carrying case equivalent to that shown on Drawing 400-533998.

3.10 Interchangeability.- All optical and mechanical parts of each azimuth and bearing circle (except for bearing ring, balancing weight and nameplates) shall be interchangeable with the same parts of any other azimuth or bearing circle of the same Mark and Mod. Selective assembly which results in noninterchangeability will not be acceptable.

3.11 Manuals.- When specified (see 6.2), a type II manual in accordance with MIL-M-15071 shall be furnished. Contents of the manuals shall be similar to Publication 0924-004-7000.

3.12 Repair parts.- When specified (see 6.2), stock repair parts shall be furnished in accordance with MIL-P-15137. Onboard repair parts shall not be furnished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in table I.

Table I - Preproduction inspection.

Examination and tests	Requirement reference	Test reference
General examination	MIL-I-983	MIL-I-983
Nonmagnetic materials	3.4.1	4.4.1
Temperature and humidity	3.5	MIL-I-983
Salt spray	3.6	MIL-I-983
Vibration	3.7	4.4.2.1
Shock	3.7	4.4.3
Alignment	3.8	4.4.4

4.2.1 Test reports.- Data obtained during the tests of 4.2 shall be recorded. One copy of such test data shall be furnished the cognizant Government inspector and one copy shall be included with the preproduction model at the time of its submission for approval to the command or agency concerned. The cover sheet of both copies of the test data shall be marked with the contract number and date.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

4.3 Quality conformance inspection. -

4.3.1 Group A examination and tests. - All equipment offered for delivery on the contract or order shall be subjected to the group A examination and tests specified in table II. Any equipment failing to meet the requirements of this specification shall not be offered for delivery.

Table II - Group A examination and tests.

Examination and tests	Requirement reference	Test reference
General examination	MIL-I-983	MIL-I-983
Nonmagnetic Materials	3.4.1	4.4.1
Alignment	3.8	4.4.4

4.3.2 Group B tests. - Group B tests, specified in table III, shall be conducted on one complete equipment when the basic design of the equipment or the material of a vital part has been changed.

Table III - Group B tests.

Tests	Requirement paragraph	Test reference
Temperature and humidity	3.5	MIL-I-983
Salt spray	3.6	MIL-I-983
Vibration	3.7	4.4.2.2
Shock	3.7	4.4.3

4.4 Test procedures. -

4.4.1 Nonmagnetic materials. - Each azimuth or bearing circle shall be placed on the bezel of an appropriate size magnetic compass and the deviation of the compass card noted. Any deflection of the compass card greater than one-half degree of arc shall be cause for rejection. Compass shall be Government furnished. In addition, all of the metallic components of the circles should be tested with a permeability meter to determine that they have a magnetic permeability less than 1.02.

4.4.2 Vibration. -

4.4.2.1 Preproduction test. - Type I vibration tests in accordance with MIL-STD-167 shall be conducted on all preproduction models.

4.4.2.2 Group B test. - The azimuth or bearing circle shall be vibrated for 10 minutes along each of the three principal axes at a frequency of approximately 33 cycles per second at an amplitude of 0.010 inch (one-half total excursion).

4.4.3 Shock. - Azimuth or bearing circles, when mounted on the bezel of a type "B" ship's course indicator (or facsimile), shall be subjected to type A HI shock tests for lightweight equipment in accordance with MIL-S-901. Release during shock of locking or detent devices shall be considered cause for rejection. These tests shall be made after all other tests have been concluded. Equipments which have been subjected to shock test shall not be put into stock nor installed onboard ship without specific written approval of the command or agency concerned.

4.4.4 Alignment. - The alignment of the assembled azimuth or bearing circle shall be tested using a collimator (see figure 3) consisting of a dummy table, representing the bezel of the magnetic compass or ship's course indicator and an artificial sun mounted at a mean angle of 45 degrees which is aligned with a

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

reference line (dumb-line) accurately inscribed on the diameter of the table. An adapter ring, 2-3/8 inches high with a wall thickness of 1/2 inch shall be used with the table so that collimation can be checked in a position simulating the use of the circles on those models of ship's course indicators in which the compass card is 2-3/8 inches (maximum) below the bezel. The outside diameter of the adapter ring and table shall be machined in accordance with Drawing S2407-533624 in order to clamp the circles in place in the same manner as in actual use.

4.4.4.1 Attach the azimuth or bearing circle to the dummy table with the plane of circle ring level and shown to be so by the spirit level. With this condition true for all angular settings of the vanes and mirrors, the line of sight through the sighting vanes, the reflected image from the dark mirror and the reflected image of the reticle line on the pentaprism shall be parallel to and coincident with the dumb-line. Failure to obtain alignment with the dumb-line will be cause for rejection.

4.4.4.2 Remove the azimuth circle from the table after completion of test specified in 4.4.4.1 and mount the table adapter, with azimuth circle attached, to the dummy table with the circle ring level and shown to be so by the spirit level. The image of the sun reflected from the cylindrical mirror through the right angle prism shall track in coincidence with the dumb-line when the mirror housing is tilted and this image shall be centered on the entrance aperture of the right angle prism to reflect a narrow beam of light in coincidence with the dumb-line directly below the right angle prism assembly. Failure to obtain alignment with the dumb-line will be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of repair parts. -

5.1.1 Basic equipment or item. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Repair parts. - Repair parts shall be prepared for the level of shipment specified (see 6.2) in accordance with MIL-0-16898.

5.1.3 Manuals. - Manuals shall be prepared for the level of shipment specified (see 6.2) in accordance with MIL-M-15071.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. - The azimuth and bearing circle covered by this specification is intended for shipboard service where it is expected to withstand frequent operation for long periods of time without benefit of frequent overhaul, except for occasional alignment inspection. The equipment enables measurement of compass bearing by directing the line of sight of a terrestrial or celestial body across the center of the compass card while maintaining the vertical axis of the compass card at right angles to the line of sight.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

(b) Quantity of azimuth or bearing circles required.
(c) If preproduction model is not required (see 3.1).
(d) Quantity of manuals required (see 3.11).
(e) Inventory control point for repair parts.
(f) Quantity of repair parts required (see 3.12).
(g) Level of preservation, packaging and packing for repair parts and manuals (see 5.1.2. and 5.1.3).
(h) Level of preservation, packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.3 Preproduction. - Invitation for bids should provide that the Government reserves the right to waive the requirement for preproduction samples to those bidders offering a product which has been previously procured and accepted by the Government as meeting this specification and that bidders offering such products, who wish to rely on such acceptance must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 6605-N104SH)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

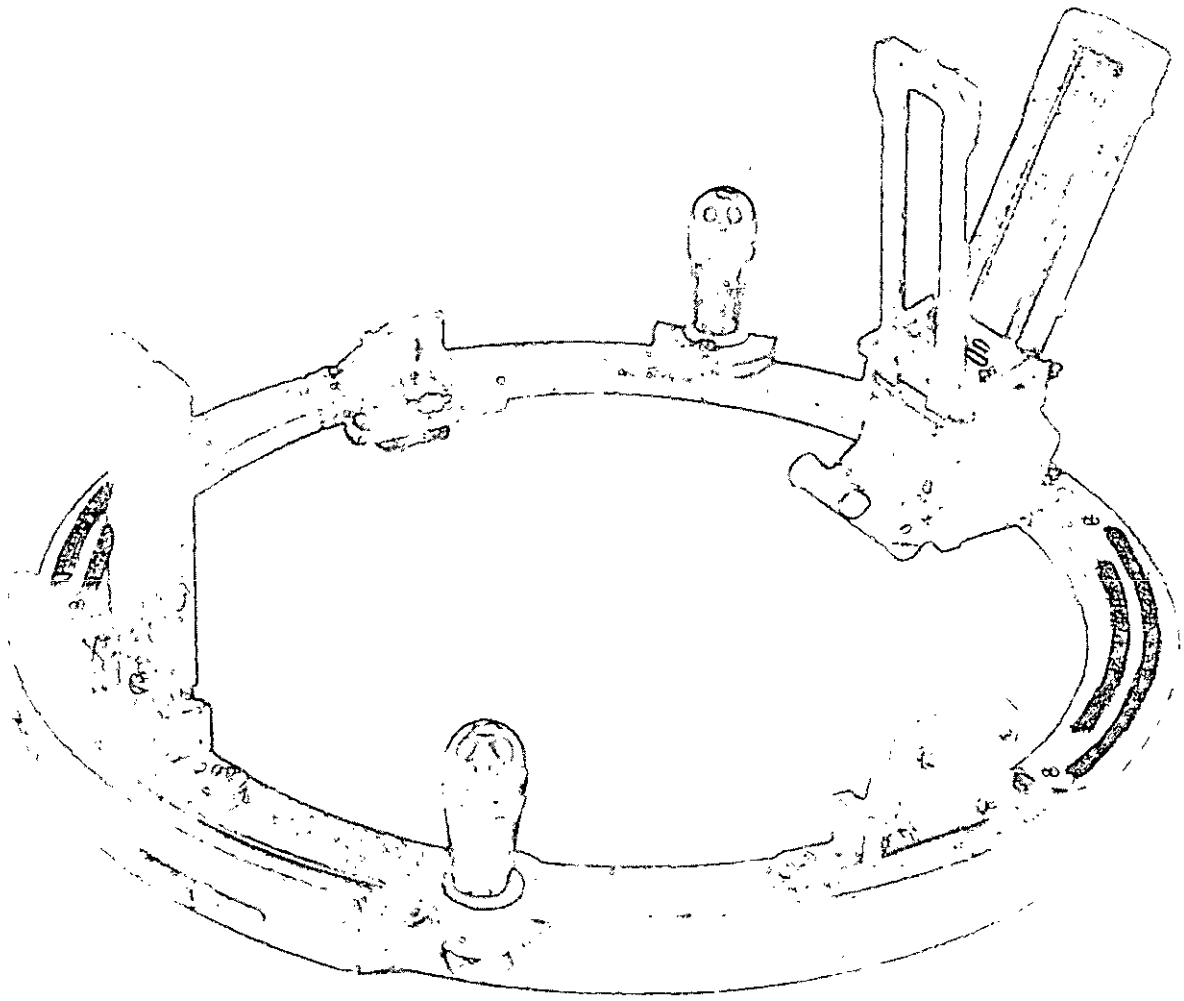


Figure 1 - General arrangement of azimuth circle, mark 4 mod 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

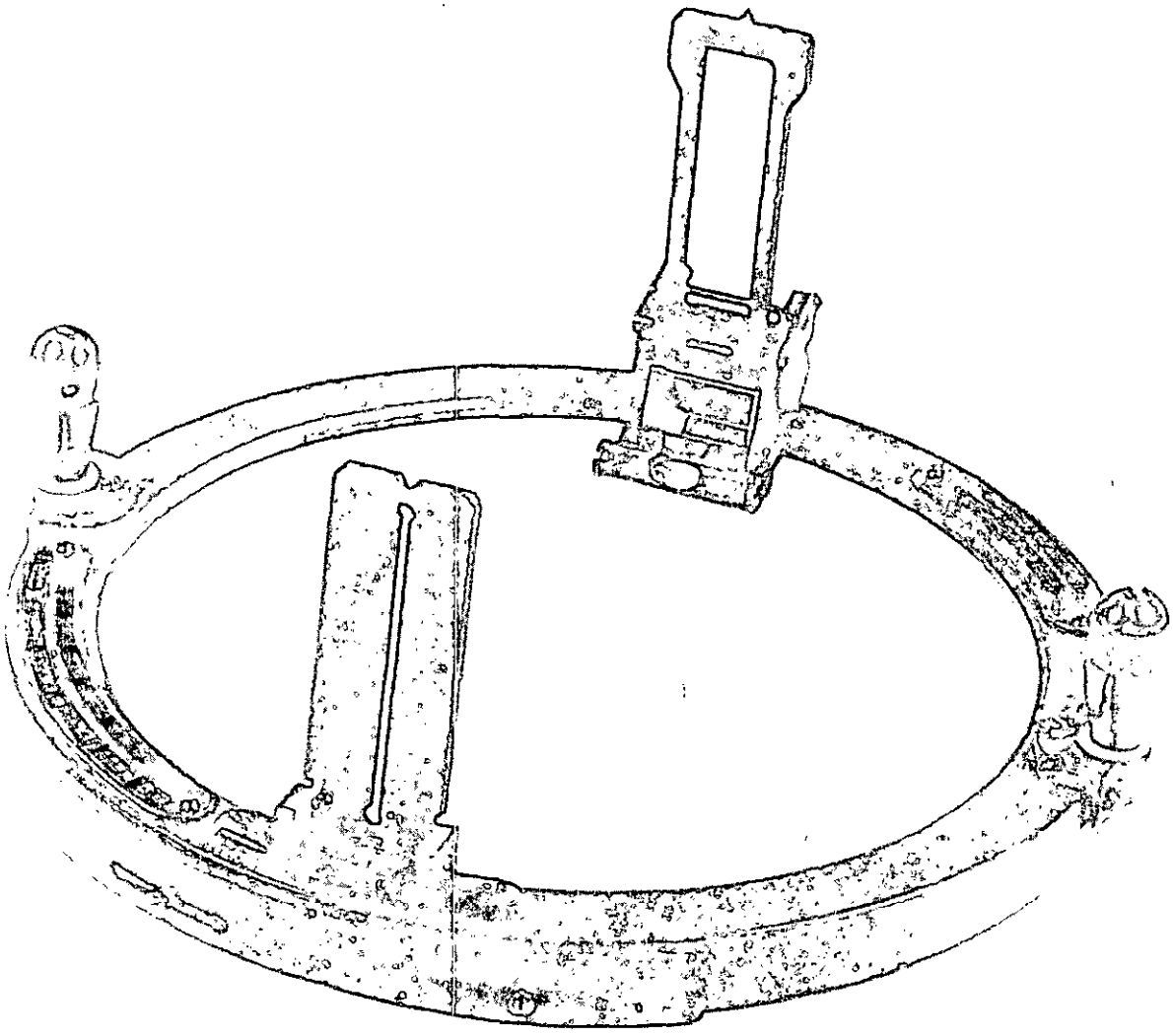


Figure 2 - General arrangement of bearing circle, mark 2 mod 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

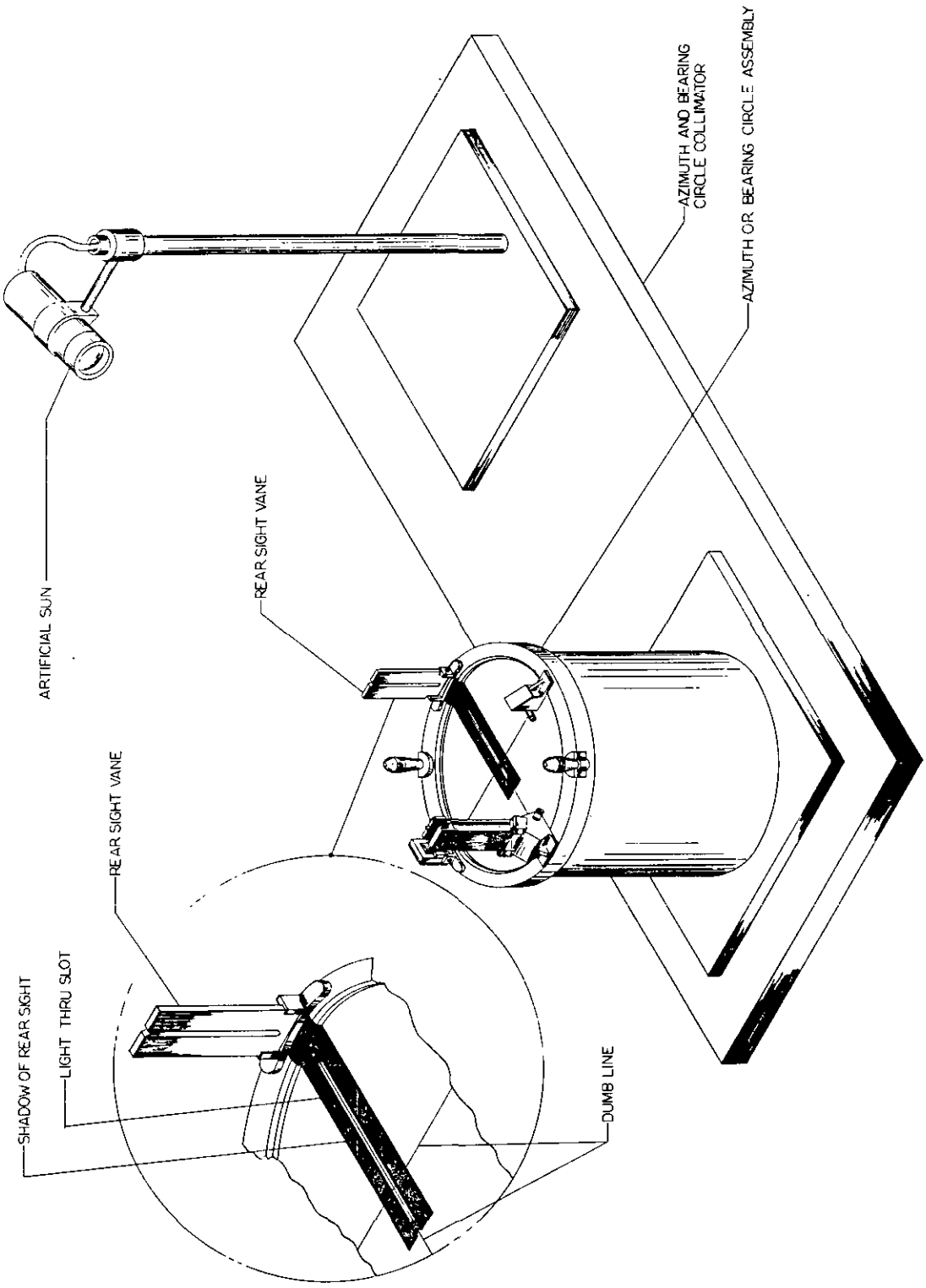


Figure 3 - Azimuth and bearing circle collimator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Ship Engineering Center
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER, NAVAL SHIP ENGINEERING CENTER
SPECIFICATIONS AND STANDARDIZATION DIVISION
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:09Z
Check the source to verify that this is the current version before use.